Cosmology Notes

Theory, Observations and Methods

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Chapter 3 Lensing

Introduction

- ***Modern Cosmology*** *Chapter 13*
- *Introduction to weak gravitational lensing*
- *Weak Gravitational Lensing*

Lensing in cosmology is a phenomenon caused by gravity bending the path of light as it travels through the universe. Gravitational lensing gives us a powerful tool to probe the distribution of matter – **not only galaxy (or baryonic) matter but also dark matter** – in the universe. We can observe lensing effects in the distorted shapes of galaxies (weak lensing), the formation of multiple images of distant objects (strong lensing), and the anisotropies in CMB.

3.1 Gravitational Lensing

Before theoretically calculating the gravitational lensing effects, we can firstly introduce three significant examples or applications of lensing:

1. **Time Delay**: Light rays emitted from the same source in the same time can be detected from different directions due to the lensing effect at different times. The delay turns out to depend on Hubble constant H_0, i.e. lensing can be used to measure H_0.
2. **Magnification**: When a compact object (e.g. a star) passes in front of a background source, it can cause a temporary increase in the brightness of the source due to lensing, which is called microlensing. This effect is used to constrain the contribution of massive compact halo objects (MACHOs) to the dark matter.

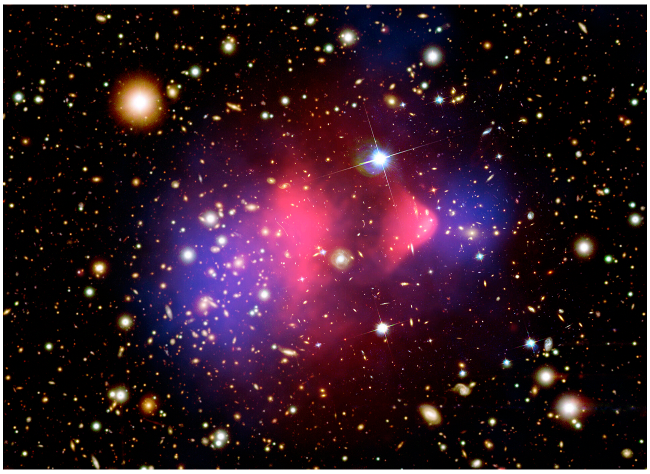

Figure 3.1: The bullet cluster, the mass distribution from lensing is shown as blue, while the X-ray emission from hot gas is shown as pink.

   Another example of magnification is the bullet cluster, which has two different possible distribution of mass from the lensing effect and from the X-ray emission. This provides a strong evidence for the existence of collisionless dark matter.
3. **Shape distortion**: The most important aspect of lensing is weak lensing, which causes slight distortions in the shapes of background galaxies. By statistically analyzing these distortions, we can map the distribution of dark matter and study the large-scale structure of the universe.

3.1.1 Weak Lensing

3.1.1.1 Describe Galaxy

To treat the lensing effect on galaxy distribution, it's hard to remap the galaxy from the observed to the true position. Instead, we can directly calculate the lensing effect on the galaxy shape, which is called weak lensing.

In the simplest case, we **statistically**[1] think the galaxy shape as circular, and the lensing effect will distort it into an ellipse. So the important part becomes quantify galaxy shape. Since lensing is the integral effect shown in the 2D spherical sky coordinate $\boldsymbol{\theta} = (\theta_1, \theta_2)$, we can use the set of second moments of its image as the simplest measurement:

$$q_{ij} = \langle \theta_i \theta_j \rangle_{I^{\rm obs}} = \frac{\int d^2\theta I(\boldsymbol{\theta}) \theta_i \theta_j}{\int d^2\theta I(\boldsymbol{\theta})}, \tag{3.1}$$

here we choose the origin such that $\langle \boldsymbol{\theta} \rangle_{I_{\rm obs}} = 0$ where $\boldsymbol{\theta}$ is the image position. Since q_{ij} is a symmetric 2×2 matrix, we can write it as:

$$q_{ij} = \frac{1}{2} q \begin{pmatrix} 1+\epsilon_1 & \epsilon_2 \\ \epsilon_2 & 1-\epsilon_1 \end{pmatrix}, \tag{3.2}$$

where $q = \mathrm{Tr}[q_{ij}]$ refers to the size of the galaxy, and ϵ_1 and ϵ_2 refer to the shape distortion, with circular shape corresponding to $\epsilon_1 = \epsilon_2 = 0$.

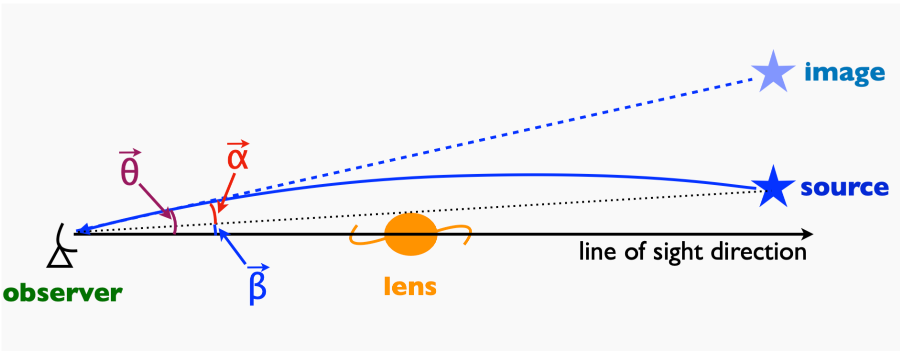

Figure 3.2: Coordinates to describe the galaxy position.

Then let us derive the effect of lensing on the galaxy shape. If we denote the true galaxy position as $\boldsymbol{\beta}$, and the deflection angel caused by mass distribution is $\boldsymbol{\alpha}(\boldsymbol{\theta})$, i.e.

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}). \tag{3.3}$$

In that case, we can describe the shape distortion by describe a galaxy with true scale $\delta\boldsymbol{\beta}$ and observed scale $\delta\boldsymbol{\theta}$:

$$\begin{cases} \boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}) \\ \boldsymbol{\beta} + \delta\boldsymbol{\beta} = \theta + \delta\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \end{cases} \Rightarrow \delta\boldsymbol{\beta} = \mathbf{A}\delta\boldsymbol{\theta}, \tag{3.4}$$

where the de-lensing matrix A_{ij} is

$$A_{ij} = \delta_{ij} - \frac{\partial \alpha_i}{\partial \theta_i}, \tag{3.5}$$

here we use Taylor expansion for $\boldsymbol{\alpha}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})$ and ignore $\mathcal{O}\left((\delta\boldsymbol{\theta})^2\right)$ since galaxy images are always very small.

[1] It's a great important idea that we can only probe weak lensing via statistics.

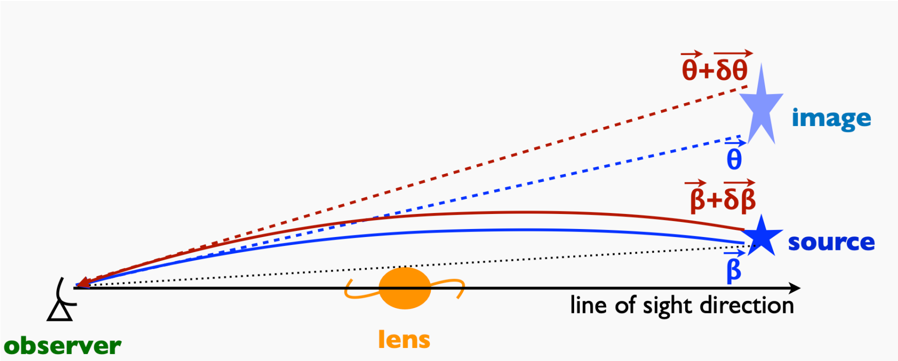

Figure 3.3: The shear from $\delta\boldsymbol{\beta}$ to $\delta\boldsymbol{\theta}$ caused by lens.

3.1.1.2 Contribution of Gravity

Now we can consider a first-order homogeneous perturbation of the metric:

$$\mathrm{d}s^2 = -\left(1+\frac{2\Psi}{c^2}\right)c^2\mathrm{d}t^2 + a^2(t)\left(1-\frac{2\Phi}{c^2}\right)\mathrm{d}l^2. \tag{3.6}$$

It is trivial that the relation between the3D spatial coordinates and the 2D angular coordinates is:

$$\begin{cases} \boldsymbol{x}_{\mathrm{true}} = (\boldsymbol{\beta}, 1)\chi, \\ \boldsymbol{x}_{\mathrm{obs}} = (\boldsymbol{\theta}, 0)\chi, \end{cases} \tag{3.7}$$

here $\boldsymbol{\theta}$ and $\boldsymbol{\beta}$ are the angular coordinates of the observed and true position, respectively, same as Eq. **??**. And χ is the comoving distance. With assuming $x^3(z)$ is the line of sight direction, we can write $\chi = x^3(z)$.

Then the true angle $\boldsymbol{\theta}$ can be calculated by integrating the transverse components' (here refers to x, y) derivation of χ:

$$\theta^i = \frac{x_\perp^i}{\chi} = \frac{1}{\chi}\int \mathrm{d}\chi \frac{\mathrm{d}x_\perp^i}{\mathrm{d}\chi} = \beta^i + 2\int_0^\chi \mathrm{d}\chi' \partial_i \Phi(\boldsymbol{\theta}, \chi')\left(1 - \frac{\chi'}{\chi}\right). \tag{3.8}$$

The second equation can be derived from general relativity, and β^i is a determined integral constant. Finally, we can write the deflection angle $\boldsymbol{\alpha}$ as:

$$\boldsymbol{\alpha} = \frac{2}{c^2}\int_0^\chi \mathrm{d}\chi' \partial_i \Phi(\boldsymbol{\theta}, \chi')\left(1 - \frac{\chi'}{\chi}\right). \tag{3.9}$$

While we can also define the lensing potential ϕ_{L} as:

$$\phi_{\mathrm{L}} = \frac{2}{c^2}\int_0^\chi \frac{\mathrm{d}\chi'}{\chi}\Phi(\boldsymbol{\theta}, \chi')\left(1 - \frac{\chi'}{\chi}\right), \tag{3.10}$$

so the deflection angle can be written as $\boldsymbol{\alpha} = \dfrac{\partial \phi_{\mathrm{L}}}{\partial \boldsymbol{\theta}}$.

3.1.1.3 Thin Lens Approximation

In another way, General relativity gives the deflection of light by a mass-point M under *Born Approximation*:

$$\delta\theta = \frac{4GM}{bc^2}, \tag{3.11}$$

with b is the impact parameter. More precisely, we can write down the formal equation of $\delta\theta$ as below:

$$\delta\theta = -\frac{2}{c^2}\int \nabla_\perp \Phi \ \mathrm{d}r. \tag{3.12}$$

Now please remember the Fig. **??**, all the calculation below is under the notion is that figure.

If we think the physical scale of the lens is much smaller than the distance between observer, lens and

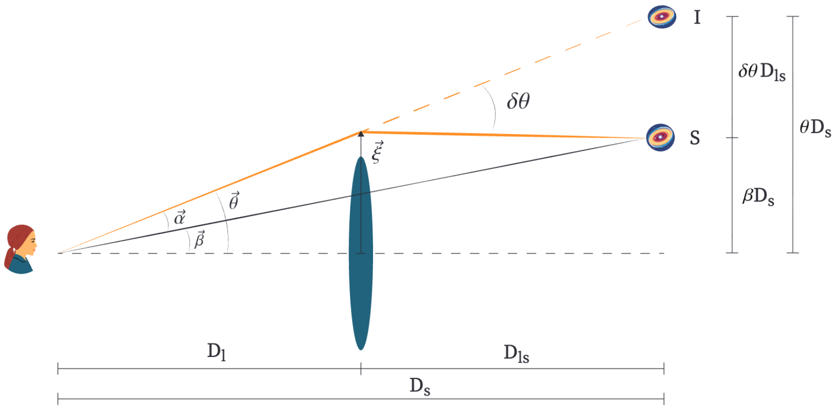

Figure 3.4: Coordinates we use.

source, then we can treat is as a **lens plane** perpendicular with the line of sight. We denote the physical position of the lens plane by $\boldsymbol{\xi}$ and call the assumption above *thin lens approximation*. For a plane, we can consider its surface density:

$$\Sigma(\boldsymbol{\xi}) = \int \mathrm{d}z \rho(\boldsymbol{\xi}, z), \tag{3.13}$$

here z is the position in the line of sight.

We can adapt the thin lens approximation with Born Approximation:

$$\mathrm{d}M = \Sigma(\boldsymbol{\xi})\mathrm{d}^2\xi, \quad b = \left|\boldsymbol{\xi} - \boldsymbol{\xi}'\right|,$$

and the direction vector $\dfrac{\boldsymbol{\xi} - \boldsymbol{\xi}'}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|}$, we finally get:

$$\delta\boldsymbol{\theta} = \frac{4G}{c^2} \int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')\Sigma(\boldsymbol{\xi}')}{\left|\boldsymbol{\xi} - \boldsymbol{\xi}'\right|^2} \, \mathrm{d}^2\boldsymbol{\xi}. \tag{3.14}$$

From Fig. **??**, we can geometrically calculate:

$$\boldsymbol{\alpha} = \frac{D_{\mathrm{ls}}}{D_{\mathrm{s}}}\delta\boldsymbol{\theta}, \quad \boldsymbol{\xi} = D_{\mathrm{l}}\boldsymbol{\theta}.$$

Substituting the relation above into Eq. **??**, we have:

$$\boldsymbol{\alpha} = \frac{4G}{c^2}\frac{D_{\mathrm{ls}}}{D_{\mathrm{l}}D_{\mathrm{s}}} \int \frac{(\boldsymbol{\theta} - \boldsymbol{\theta}')\Sigma(D_{\mathrm{l}}\boldsymbol{\theta}')}{\left|\boldsymbol{\theta} - \boldsymbol{\theta}'\right|^2} \, \mathrm{d}^2\boldsymbol{\theta} = \frac{1}{\pi} \int \kappa(\boldsymbol{\theta}) \frac{(\boldsymbol{\theta} - \boldsymbol{\theta}')}{\left|\boldsymbol{\theta} - \boldsymbol{\theta}'\right|^2} \, \mathrm{d}^2\theta. \tag{3.15}$$

Here $\kappa(\boldsymbol{\theta})$ is the **convergence field**, referring to the dimensionless surface mass density of lens:

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(\boldsymbol{\theta})}{\Sigma_{\mathrm{cirt}}}, \quad \Sigma_{\mathrm{cirt}} = \frac{c^2}{4\pi G}\frac{D_{\mathrm{os}}}{D_{\mathrm{l}}D_{\mathrm{s}}}. \tag{3.16}$$

Also, substituting the relation to Eq. **??**, we can derive the **lensing potential** ϕ_{L}:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{\partial}{\partial\boldsymbol{\theta}}\phi_{\mathrm{L}} = \frac{\partial}{\partial\boldsymbol{\theta}}\left[\frac{2}{c^2}\frac{D_{\mathrm{ls}}}{D_{\mathrm{l}}D_{\mathrm{s}}} \int \Phi(D_{\mathrm{l}}\boldsymbol{\theta}, z) \, \mathrm{d}z\right]. \tag{3.17}$$

This equation is consistent with the Eq. **??** we derived from the first-order homogeneous perturbation of the metric, with $\dfrac{D_{\mathrm{ls}}}{D_{\mathrm{l}}D_{\mathrm{s}}} \to \dfrac{\chi - \chi'}{\chi\chi'}$.

3.1.1.4 Convergence and Shear

Now review Eq. **??**, the trace of $\mathbf{A}$ is

$$\mathrm{Tr}[A_{ij}] = 2 - \frac{\partial \alpha_1}{\partial \theta_1} - \frac{\partial \alpha_2}{\partial \theta_2} = 2 - \kappa(\boldsymbol{\theta}),$$

so we can rewrite A_{ij}:

$$A_{ij} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}. \tag{3.18}$$

3.1.1.5 Convergence as the projected matter density

From the definition of lensing potential ϕ_{L}, we can derive the convergence κ as:

$$\kappa = \frac{1}{2}\frac{\partial^2 \phi_{\mathrm{L}}}{\partial \theta^2} = \frac{1}{c^2}\int_0^{\chi} \mathrm{d}\chi' \frac{\chi'(\chi-\chi')}{\chi}\nabla^2\Phi(\boldsymbol{\theta},\chi') = \frac{3H_0^2\Omega_{\mathrm{m}}}{2c^2}\int_0^{\chi}\mathrm{d}\chi' \frac{\chi'(\chi-\chi')}{\chi}\frac{\delta(\boldsymbol{\theta},\chi')}{a}. \tag{3.19}$$

This equation shows that the convergence κ is essentially a weighted projection of the matter density contrast δ along the line of sight.

3.1.2 Strong Lensing

3.2 CMB Lensing

Chapter 4 Other Probes

4.1 Cosmic Chronometer

4.2 $21\,\mathrm{cm}$ Cosmology

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Part V

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7.2 Neutrino Mass and Hierarchy

7.3 Non-Gaussianity

Introduction

- *Acoustic Signatures in the Primary Microwave Background Bispectrum*
- *Squeezing $f_{\rm NL}$ out of the matter bispectrum with consistency relations*
- *Stochastic Bias from Non-Gaussian Initial Conditions*

7.3.1 Statistical Imprint of Non-Gaussianity

In this section, we mainly focus on the local-type non-Gaussianity and its imprint in statistical measurements, in contrast of primordial non-Gaussianity during inflation. The non-Gaussianity can be parameterized as:

$$\Phi(\boldsymbol{x}) = \phi(\boldsymbol{x}) + f_{\rm NL}(\phi^2(\boldsymbol{x}) - \langle\phi^2\rangle), \tag{7.1}$$

where Φ is the Bardeen potential, ϕ is a Gaussian random field, and $f_{\rm NL}$ is the non-Gaussianity parameter. And its Fourier transform is:

$$\Phi(\boldsymbol{k}) = \phi(\boldsymbol{k}) + f_{\rm NL}\varphi(\boldsymbol{k}). \tag{7.2}$$

Here $\varphi(\boldsymbol{k})$ is the non-linear part of the potential, which is a convolution of ϕ:

$$\varphi(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 k_1}{(2\pi)^3}\phi(\boldsymbol{k}_1)\phi(\boldsymbol{k}-\boldsymbol{k}_1) - (2\pi)^3\delta(\boldsymbol{k})\langle\phi^2\rangle. \tag{7.3}$$

Furthermore, we can write down:

$$\phi(\boldsymbol{x}) = \phi_{\rm L}(\boldsymbol{x}) + \phi_{\rm S}(\boldsymbol{x}), \tag{7.4}$$

as we divided the gaussian field into **two independent part** – long-wavelength part and short-wavelength part. Then substitute it into Eq. **??**, we can get:

$$\Phi(\boldsymbol{x}) = \phi_{\rm L}(\boldsymbol{x}) + f_{\rm NL}(\phi_{\rm L}^2(\boldsymbol{x}) - \langle\phi_{\rm L}^2\rangle) + (1 + 2f_{\rm NL}\phi_{\rm L}(\boldsymbol{x}))\phi_{\rm S}(\boldsymbol{x}) + f_{\rm NL}(\phi_{\rm S}^2(\boldsymbol{x}) - \langle\phi_{\rm S}^2\rangle). \tag{7.5}$$

7.3.1.1 Squeezed Limit of the Bispectrum

One way to detect $f_{\rm NL}$ is to measure the bispectrum $B(k_1, k_2, k_3)$. The bispectrum of the potential Φ can be calculated as:

$$\begin{aligned} B_\Phi(k_1, k_2, k_3) &= \langle \Phi(k_1)\Phi(k_2)\Phi(k_3)\rangle \\ &= \langle (\phi(k_1) + f_{\rm NL}\varphi(k_1))(\phi(k_2) + f_{\rm NL}(\varphi(k_2)))(\phi(k_3) + f_{\rm NL}(\varphi(k_3)))\rangle \\ &\simeq \langle \phi(k_1)\phi(k_2)\phi(k_3)\rangle + f_{\rm NL}\langle \phi(k_1)\phi(k_2)\varphi(k_3)\rangle + f_{\rm NL}\langle \phi(k_1)\varphi(k_2)\phi(k_3)\rangle + f_{\rm NL}\langle \varphi(k_1)\phi(k_2)\phi(k_3)\rangle \\ &= 0 + f_{\rm NL}A_{ij,k} + \text{permutations} \end{aligned}$$

where $\simeq$ means we ignore the higher-order terms of $f_{\rm NL}$, and $A_{ij,k} = \langle \phi(k_i)\phi(k_j)\varphi(k_k)\rangle$. We can further calculate $A_{ij,k}$ as:

$$\begin{aligned} A_{ij,k} &= \int \frac{\mathrm{d}^3k'}{(2\pi)^3}\langle \phi(k_i)\phi(k_j)\phi(k')\phi(k_k - k')\rangle - (2\pi)^3\delta(k_k)\langle\phi^2\rangle\langle\phi(k_i)\phi(k_j)\rangle \\ &= 2\int \frac{\mathrm{d}^3k'}{(2\pi)^3}P_\phi(k_i)P_\phi(k_j)\delta(\boldsymbol{k}_i + \boldsymbol{k}_j)\delta(\boldsymbol{k}_i + \boldsymbol{k}')\delta(\boldsymbol{k}_j + \boldsymbol{k}_k - \boldsymbol{k}') \\ &= 2P_\phi(k_i)P_\phi(k_j)\delta(\boldsymbol{k}_i + \boldsymbol{k}_j + \boldsymbol{k}_k). \end{aligned}$$

The second equation is expanded by Wick theorem. Finally, we can write the bispectrum of Φ as:

$$B_\Phi(k_1, k_2, k_3) = 2f_{\rm NL}[P_\phi(k_1)P_\phi(k_2) + P_\phi(k_2)P_\phi(k_3) + P_\phi(k_3)P_\phi(k_1)]\delta(\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3). \tag{7.6}$$

Squeezed limit of the bispectrum means $q \equiv k_1 \ll k_2 \approx k_3 \equiv k$, so the bispectrum in this limit can be written as:

$$B_\Phi(q, k, k) \simeq 4f_{\rm NL}P_\phi(q)P_\phi(k) + \mathscr{O}(q). \tag{7.7}$$

In observation, we cannot directly measure the bispectrum of the potential Φ, but we can measure the bispectrum of the matter density contrast δ using Poisson equation:

$$\delta(k) = \frac{2}{3}\frac{k^2T(k)D(z)}{\Omega_{\rm m}H_0^2}\Phi(k) \equiv \alpha(k, z)\Phi(k). \tag{7.8}$$

So the squeezed limit of the bispectrum of δ can be written as:

$$B_\delta(q, k, k) \simeq 4f_{\rm NL}P_\phi(q)P_\phi(k)\alpha(q)\alpha^2(k) = 4f_{\rm NL}\frac{P_\delta(q)P_\delta(k)}{\alpha(q)}. \tag{7.9}$$

Since $\alpha(q) \propto q^2$ at large scale, we often generally write the squeezed limit with remainder term as:

$$B_\delta(q, k, k) = a_{f_{\rm NL}}\frac{P_\delta(q)P_\delta(k)}{q^2T(q)} + \sum_{n=0}^{\infty} a_n\left(\frac{q}{k}\right)P_\delta(q)P_\delta(k). \tag{7.10}$$

After this, we leave out the sub-script δ for simplicity. A more general form of the squeezed limit is derived in Squeezing $f_{\rm NL}$ out of the matter bispectrum with consistency relations, which gives $a_{f_{\rm NL}}$ as:

$$a_{f_{\rm NL}} = \frac{6f_{\rm NL}\Omega_{\rm m}H_0^2}{D(z)}\frac{\partial \log P(k)}{\partial \log \sigma_8}. \tag{7.11}$$

Here we do not present the detailed derivation but point out the key equation of squeezed limit:

$$B(q, k) = V\langle \delta(q)P(k|\Phi_{\rm L})\rangle = V\sum_{q'}\frac{\partial P(k)}{\partial \Phi_{\rm L}(q')}\langle \delta_q\Phi_{\rm L}(q')\rangle = V\frac{\partial P(k)}{\partial \Phi_{L,-q}}\frac{P(q)}{\alpha(q)} + \ldots. \tag{7.12}$$

here $V = (2\pi)^3\delta(\mathbf{0})$ and the second equation is the expansion of $P(k|\Phi_{\rm L})$ in terms of long-wavelength (soft model) ϕ as the background.

7.3.1.2 Scale-dependent Bias

Another more simple imprint caused by non-Gaussianity is the scale-dependent bias, with the same idea of squeezed limit that long-wavelength mode (soft mode) modulate short-wavelength mode (hard mode).

Consider local statistical $\sigma_8(\boldsymbol{x})$, if we ignore the background (soft mode), we can write σ_8 as:

$$\sigma_8(\boldsymbol{x}) = (1 + 2f_{\mathrm{NL}}\phi_{\mathrm{L}})\bar{\sigma}_8. \tag{7.13}$$

This is derived from Eq. **??**. Then the halo bias can be written as:

$$\delta_{\mathrm{h}} \equiv \frac{\delta n_{\mathrm{h}}}{\bar{n}_{\mathrm{h}}} = \frac{\partial \ln n_{\mathrm{h}}}{\partial \delta}\delta + \frac{\partial \ln n_{\mathrm{h}}}{\partial \ln \sigma_8}\frac{\delta \sigma_8}{\sigma_8}. \tag{7.14}$$

Suppose $b_{\mathrm{h}} = \dfrac{\partial \ln n_{\mathrm{h}}}{\partial \delta_{\mathrm{L}}}$ and $\beta_{\mathrm{h}} = 2\dfrac{\partial \ln n_{\mathrm{h}}}{\partial \ln \sigma_8}$ are the Gaussian bias and non-Gaussian bias, we can write the halo bias as:

$$\delta_{\mathrm{h}} = \left[b_{\mathrm{h}} + \beta_{\mathrm{h}}\frac{f_{\mathrm{NL}}}{\alpha}\right]\delta_{\mathrm{L}}, \tag{7.15}$$

which suits for the large scale limit.

Stochastic Bias from Non-Gaussian Initial Conditions shows that the equation above is not exact if the inflation is not a single field, and there is an extra stochastic bias term ϵ for a general conclusion:

$$\delta_{\mathrm{h}} = \left[b_{\mathrm{h}} + \beta_{\mathrm{h}}\frac{f_{\mathrm{NL}}}{\alpha}\right]\delta_{\mathrm{L}} + \epsilon. \tag{7.16}$$

7.3.2 Primordial Non-Gaussianity during Inflation

Chapter 8 Reconstructing the Universe

8.1 Reconstruction of Density Field

8.2 Reconstruction of Velocity Field

8.3 Reconstruction of Initial Conditions

Appendix A Math for Sphere Projection

Introduction

- *Three-dimensional Fourier transforms, integrals of spherical Bessel functions, and novel delta function identities*

A.1 Fourier Transform in Spherical Coordinates

For an isotropic function $f(\boldsymbol{r}) = f(r)$, the Fourier transform can be simplified as:

$$\begin{aligned}
\tilde{f}(\boldsymbol{k}) &= \int \mathrm{d}^3 r f(\boldsymbol{r}) e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} \\
&= \int_0^\infty r^2 \mathrm{d}r f(r) \int_0^\pi \sin\theta \mathrm{d}\theta e^{-ikr\cos\theta} \int_0^{2\pi} \mathrm{d}\phi \\
&= 2\pi \int_0^\infty r^2 \mathrm{d}r f(r) \int_{-1}^1 \mathrm{d}(\cos\theta)\, e^{-ikr\cos\theta} \\
&= 4\pi \int_0^\infty r^2 \mathrm{d}r f(r) \frac{\sin(kr)}{kr} \\
&= 4\pi \int_0^\infty r^2 \mathrm{d}r f(r) j_0(kr),
\end{aligned}$$

where $j_0(kr) = \sin(kr)/(kr)$ is the spherical Bessel function of order 0. The inverse Fourier transform can be written as:

$$\begin{aligned}
f(\boldsymbol{r}) &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \tilde{f}(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \\
&= \int_0^\infty \frac{k^2 \mathrm{d}k}{(2\pi)^3} \tilde{f}(k) \int_0^\pi \sin\theta \mathrm{d}\theta e^{ikr\cos\theta} \int_0^{2\pi} \mathrm{d}\phi \\
&= \frac{1}{2\pi^2} \int_0^\infty k^2 \mathrm{d}k \tilde{f}(k) \frac{\sin(kr)}{kr} \\
&= \frac{1}{2\pi^2} \int_0^\infty k^2 \mathrm{d}k \tilde{f}(k) j_0(kr).
\end{aligned}$$